Materials Characterization 229 (2025) 115480

Contents lists available at ScienceDirect

Materials Characterization

journal homepage: www.elsevier.com/locate/matchar

Making low alloy ultrahigh strength steel tough by high-density NiAl clusters

Xiaohan Duan [a,b], Xin Cao [c], Yangxin Wang [a,b,c,*], Hongshan Zhao [a,b,c], Chundong Hu [a,b,c,*], Han Dong [a,b,c]

[a] State Key Laboratory of Materials for Advanced Nuclear Energy, Shanghai University, Shanghai 200444, China
[b] School of Materials Science and Engineering, Shanghai University, Shanghai 200444, China
[c] Zhejiang Institute of Advanced Materials, SHU, Jiashan 314100, China

ARTICLE INFO

Keywords:
Low alloy ultrahigh strength steel
NiAl clusters
Dual-precipitation
Toughening mechanisms

ABSTRACT

A novel low alloy ultrahigh strength steel was developed by introducing extremely high-density NiAl clusters (~ 4.88×10^{24} m^{-3}). The steel exhibited an ultimate tensile strength of ~ 1.9 GPa, a total elongation of ~ 15 %, a U-notch impact absorption energy of ~ 80 J and a fracture toughness of 108 MPa·$m^{1/2}$. The precipitation behavior and the toughening mechanisms were systematically investigated. Highly dispersed nano-NiAl clusters and submicron ε-carbides precipitate independently, although a small fraction of NiAl clusters interacts with Si during precipitation, thereby increasing the availability of heterogeneous nucleation sites. NiAl clusters promote uniform dislocation distribution, suppressing pile-ups at boundaries and delaying crack initiation. The study revealed multi-scale toughening mechanisms, including large dimple growth, uniform plastic deformation, secondary crack formation, and extensive crack deflection, which collectively result in a ~ 23 % improvement in impact absorption energy. These findings demonstrate that dual-precipitation of NiAl cluster and ε-carbides and dislocation-precipitate interactions provides an effective strategy for achieving high toughness at a tensile strength of ~ 1.9 GPa in low alloy ultrahigh strength steels.

1. Introduction

Low alloy ultrahigh strength steels, such as 300 M, D6AC, and Eglin, are commonly used in aircraft landing gear, rocket casings, and heavy-duty equipment owing to their excellent strength-toughness combination and cost efficiency [1–3]. Upon quenching and tempering, these steels achieve tensile strengths exceeding 1.8 GPa through a combination of grain refinement by MC carbides, dislocation strengthening within the martensitic matrix, and further enhancement by the precipitation of submicron-sized ε-carbides during low-temperature tempering [4–6].

300 M is among the highest-strength low alloy steels currently available. Mandal et al. [6] reported that 300 M tempered at 850 °C followed by water quenching achieved a tensile strength of 1991 MPa, but exhibited a limited V-notch impact absorption energy of only 27.6 J. With increasing tensile strength, a reduction in impact toughness is commonly observed, reflecting the inherent strength-toughness trade-off that continues to constrain the structural reliability of such steels [7–9].

In efforts to enhance the toughness of low-alloy steels, considerable attention has been directed toward microstructural optimization by controlling high-angle boundaries [10] and tailoring the amount and distribution of retained austenite [5,11].However, these approaches typically rely on deformation or cryogenic treatments, which involve complex procedures and offer limited process control. In contrast, the introduction of nanoscale precipitates has emerged as a more favorable strategy in recent years, offering a simultaneous improvement in strength and toughness. Although conventional carbides such as M_2C [12] and MC [13] can provide effective strengthening, their formation depends on C, which may partially degrade the matrix. To overcome this limitation, increasing attention has been directed toward intermetallic compounds-particularly nanoscale NiAl precipitates-which are capable of delivering substantial strengthening [14,15] under medium-low-carbon conditions while maintaining good ductility making them one of the most promising directions in current alloy design [16,17].

B2-NiAl shows exceptional potential in high-alloy maraging steels

* Corresponding authors at: School of Materials Science and Engineering, Shanghai University, Shanghai 200444, China.
E-mail addresses: wangyangxin81@shu.edu.cn (Y. Wang), hcd@shu.edu.cn (C. Hu).

https://doi.org/10.1016/j.matchar.2025.115480
Received 25 June 2025; Received in revised form 10 August 2025; Accepted 13 August 2025
Available online 18 August 2025
1044-5803/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

due to its high lattice coherency with the α-Fe matrix (lattice misfit <3 %) [18], metastable nanocluster features, and dislocation shearing mechanisms. Peng et al. [19] demonstrated that the introduction of Mo-rich clusters could effectively promote the precipitation of high-density coherent NiAl phases. Benefiting from low lattice mismatch and pronounced ordering effects, the resulting maraging steel exhibited a tensile strength of 2099 MPa along with a fracture toughness exceeding 80 MPa·$m^{1/2}$. This study also offers new insights into the strength-toughness design of low-alloy steels, suggesting that the introduction of nanoscale NiAl precipitates with high number density may enhance toughness while maintaining a tensile strength near 1900 MPa.

Accordingly, a low alloy ultrahigh strength steel was designed with dual precipitation of ε-carbides and NiAl clusters with an extremely high number density. This study systematically investigates the nucleation mechanisms of NiAl and ε phases, evaluates the influence of NiAl clusters on the hierarchical martensitic microstructure, and elucidates their precipitation behavior and toughening contributions.

2. Experimental

2.1. Materials, treatment, and mechanical properties

Table 1 presents the chemical compositions of the two experimental steels, referred to as 0Al and 1.3Al for simplicity. To ensure sufficient NiAl content in the 1.3Al steel, the Ni and Al concentrations were appropriately increased based on the composition of the 0Al steel. Both materials were produced by vacuum induction melting followed by vacuum arc remelting (VIM + VAR) to obtain 500 kg ingots, which were subsequently forged into billets with Φ170 mm. Specimens for microstructural and mechanical characterization were extracted from the quarter-radius position.

The two steels underwent solution treatment at 970 °C for 1 h to dissolve carbides, followed by tempering at 180 °C for 2 h, with water quenching after each step. The samples of 0Al steel before and after tempering are denoted as 0Al,WQ and 0Al,2 h, respectively, while those of 1.3Al steel are designated as 1.3Al,WQ and 1.3Al,2 h. Room-temperature tensile tests were conducted on specimens with a diameter of 5 mm and a gauge length of 25 mm using an testing machine (Instron 5985) at a strain rate of 0.001 s^{-1}. Charpy U-notch impact toughness (KU_2) was measured on specimens with dimensions of 10 × 10 × 55 mm^3 using an impact tester (Instron 750MPX). Both the specimen surfaces and notches were mechanically polished to minimize surface roughness effects. Three tests per sample condition were performed. Plane strain fracture toughness tests were conducted on a servo-hydraulic fatigue machine (Instron 8801) using a three-point bending configuration. Specimens with a thickness of 25 mm and a width of 50 mm were machined, with a 3 mm fatigue pre-crack introduced at the notch tip. During testing, a loading rate of 38 KN/min was applied. The conditional fracture toughness value K_Q was calculated in accordance with ASTM E399 using the following equation:

$$K_Q = \left(\frac{F_Q S}{BW^{3/2}}\right) \times f\left(\frac{a}{W}\right) \quad (1)$$

where F_Q is the conditional value of the load; S, a, B, W are the loading span, crack length, thickness and width of the bend specimens, respectively; and f is the dimensional factor which depends on the ratio of a to W.

Table 1
Matrix compositions of the 0Al and 1.3Al steels. (mass fraction, wt%)

Steel	C	Si	Mn	Cr	Ni	Mo	V	Al
0Al	0.32	1.98	1.02	1.00	3.41	0.56	0.09	0.03
1.3Al	0.31	2.01	1.03	1.00	5.06	0.60	0.11	1.30

2.2. Microstructural characterization and data analysis

The microstructures of the specimens were characterized by scanning electron microscopy (SEM, Zeiss Sigma 300) and transmission electron microscopy (TEM, JEM-F200). SEM specimens were cut from impact specimens, ground, polished, and subsequently etched with a 4 vol% nitric acid-ethanol solution. Thin foils for TEM were extracted from the matrix, ground to 50 μm, punched into Φ3 mm discs, and electropolished by double-jet thinning at 30 V and − 25 °C using a 10 vol % perchloric acid alcohol solution. TEM imaging was conducted at an accelerating voltage of 200 KV. Regions near fracture surfaces were sampled by focused ion beam (FIB) milling.

Electron backscatter diffraction (EBSD) was conducted on specimens measuring 10 × 10 × 5 mm^3, which were mechanically polished with 0.02 μm colloidal silica suspension to eliminate surface residual stresses. EBSD data were collected using a Bruker e-Flash FS detector mounted on the SEM, operating at an accelerating voltage of 20 KV and a step size of 0.1 μm. Subsequent data processing was performed using Aztec Crystal v2.1 software.

Phase analysis was conducted using X-ray diffraction (XRD, Rigaku D/Max-2550) with a Co radiation source operated at 40 KV and 150 mA. Scans were performed from 20° to 110° at a scanning rate of 2°/min. The volume fraction of retained austenite was determined by the direct comparison method, assuming negligible diffraction contributions from carbides. The dislocation density (ρ) was estimated using the Williamson-Hall method [20]:

$$\frac{\Delta 2\theta \cos\theta}{\lambda} = \frac{0.9}{D} + \frac{2\varepsilon \sin\theta}{\lambda} \quad (2)$$

where λ is the X-ray wavelength (0.154 nm), D is the apparent crystallite size, and ε represents the lattice strain. The dislocation density (ρ) was calculated based on the following equation [21]:$\rho = \frac{2\sqrt{3}\varepsilon}{Db}$, where b denotes the magnitude of the Burgers vector.

Atom probe tomography (APT, LEAP 5000XR) was performed in voltage-pulsing mode. Specimens were prepared by electrochemical polishing of needle-shaped blanks and subsequently analyzed at a base temperature of 50 K under a pulse fraction of 0.2 and a repetition rate of 125 KHZ. The detection rate was controlled at 5 ions per 1 k pulses by gradually increasing the applied voltage. APT data were reconstructed and analyzed using CAMECA IVAS 3.6.8 software, and the average diameter and number density of precipitates were evaluated [22].

3. Results

3.1. Mechanical properties

Fig. 1(a) shows the engineering stress-strain curves of the two steels before and after tempering. Both steels exhibited high ultimate tensile strength (*UTS*) and yield strength (*YS*) in the as-quenched condition, reaching 2294 ± 9 MPa, 1234 ± 22 MPa for the 0Al steel and 2145 ± 144 MPa, 1113 ± 11 MPa for the 1.3Al steel, with corresponding total elongations of 9.5 ± 0.6 % and 4.3 ± 0.9 %, respectively. Notably, the 1.3Al steel fractured prior to reaching its maximum tensile strength during the tensile test. After tempering, the *UTS* and *YS* values of the 0Al and 1.3Al steels decreased to 1925 ± 5 MPa, 1400 ± 12 MPa and 1876 ± 25 MPa, 1272 ± 52 MPa respectively, both falling below 2000 MPa. However, the TE values increased significantly to 13.5 ± 1.8 % and 13.7 ± 0.4 % for the 0Al and 1.3Al steels, respectively. The detailed mechanical properties are summarized in Table 2.

To further assess the toughness variation, the KU_2 values were measured and summarized in Fig. 1(b). For the 0Al steel, the KU_2 increased from 33 ± 6 J to 65 ± 3 J after 2 h of tempering, while for the 1.3Al steel, it rose sharply from 23 ± 2 J to approximately 80 ± 3 J, corresponding to an improvement of about 3.5 times compared to its as-quenched state. The fracture toughness (K_{IC}) of the 1.3Al steel reached

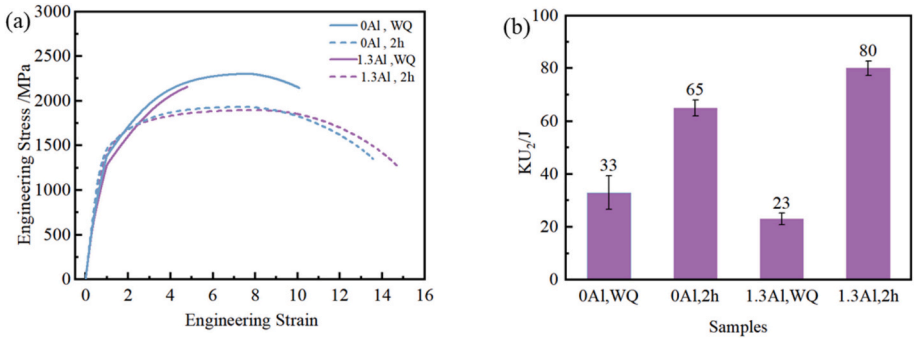


Fig. 1. Mechanical properties of the two steels before and after tempering at 180 °C: (a) engineering stress-strain curves; (b) impact absorbing energy.

Table 2
Yield strength (*YS*), Ultimate tensile strength (*UTS*), Total elongation (*TE*), impact absorption energy (KU_2) of the 0Al and 1.3Al steels in the as quenched (WQ) and 2 h tempered conditions.

Steel	Condition	*YS* (MPa)	*UTS* (MPa)	*TE* (%)	KU_2 (J)	K_{IC}(MPa·m$^{1/2}$)
0Al	WQ	1234 ± 22	2294 ± 9	9.5 ± 0.6	33 ± 6	–
	2 h	1400 ± 12	1925 ± 5	13.5 ± 1.8	65 ± 3	97.6
1.3Al	WQ	1113 ± 11	2145 ± 144	4.3 ± 0.9	23 ± 2	–
	2 h	1272 ± 52	1876 ± 25	13.7 ± 0.4	80 ± 3	108.0

108.0 MPa·m$^{1/2}$, about 10.4 MPa·m$^{1/2}$ higher than that of the 0Al counterpart (97.6 MPa·m$^{1/2}$). It is noteworthy that under comparable strength levels, the 1.3Al,2 h steel demonstrated superior toughness compared to the 0Al,2 h steel.

3.2. Microstructures

Supplementary Fig. S1 presents the microstructures of the two steels before and after tempering. Fig. 2(a) shows the microstructure of the 1.3Al steel (tempered state, hereafter referred to as 0Al and 1.3Al steels), while both the 0Al and 1.3Al steels exhibit tempered martensitic structures. XRD patterns in Fig. 2(b) reveal that a small amount of retained austenite remains in both steels after tempering. Based on the XRD results, assuming negligible diffraction contributions from carbides [10], the retained austenite contents were determined to be 1.8 % for the 1.3Al steel and 1.0 % for the 0Al steel, respectively, with a marginal difference of 0.8 %. Dislocation densities were estimated using the Williamson-Hall approach by fitting the $(110)_M$, $(200)_M$, and $(211)_M$ diffraction peaks with Jade 6.0 software [23]. The calculated dislocation densities were ~ 2.20 × 10^{15} m^{-2} for the 1.3Al steel and ~ 2.01 × 10^{15} m^{-2} for the 0Al steel, indicating a slight increase in dislocation density after Al addition.

Fig. 3 shows the morphologies of martensitic laths, retained austenite and ε-carbides in the 0Al and 1.3Al steels after tempering. The corresponding selected area electron diffraction (SAED) patterns are displayed in the upper-right corners. In Fig. 3(a) and Fig. 3(d), the matrix is composed of tempered martensitic laths with high dislocation densities, retaining sharp boundaries and well-defined structures after tempering. Thin films of retained austenite were observed at lath boundaries and identified by dark-field imaging and SAED (Fig. 3(b)). The retained austenite was indexed as $(110)_\gamma$ and the adjacent martensite as $(111)_M$, consistent with the Kurdjumov-Sachs (K—S) orientation relationship of $\{111\}_\gamma//\{110\}_M$, $\langle 110\rangle_\gamma//\langle 111\rangle_M$. As shown in Fig. 3(c) and Fig. 3(f), numerous needle-like ε-carbides with lengths of 100–500 nm and widths of 3–10 nm precipitated uniformly within the laths, aligned along a common crystallographic direction.

Prior austenite grain sizes and martensitic units (packets, blocks, and laths) in the 0Al and 1.3Al steels were measured by EBSD, SEM, and TEM, as summarized in Fig. 4 and Table 3. Average values were determined from measurements of approximately 300 prior austenite grains, 200 martensitic packets, 200 blocks, and 100 lath widths. The average prior austenite grain sizes (d_a) were 26.1 ± 9.4 μm for the 0Al steel and 23.4 ± 5.9 μm for the 1.3Al steel, showing minimal difference. Similarly, the sizes of martensitic packets (d_p), blocks (d_b), and laths (d_l), characterized by EBSD and TEM, were comparable between the two steels. These results indicate that martensitic unit size is not the primary factor governing KU_2 and K_{IC}, and suggest a dominant influence from differences in precipitation behavior.

3.3. Precipitate characterization

Fig. 5(a) and (d) show the 5 at.% C and 10 at.% Ni + Al iso-concentration surfaces in the 1.3Al steel. Needle-like C-rich precipitates, with an average thickness of ~2.2 nm and a spacing of ~10 nm, are uniformly distributed within the matrix(Fig. 5(a)).TEM observations

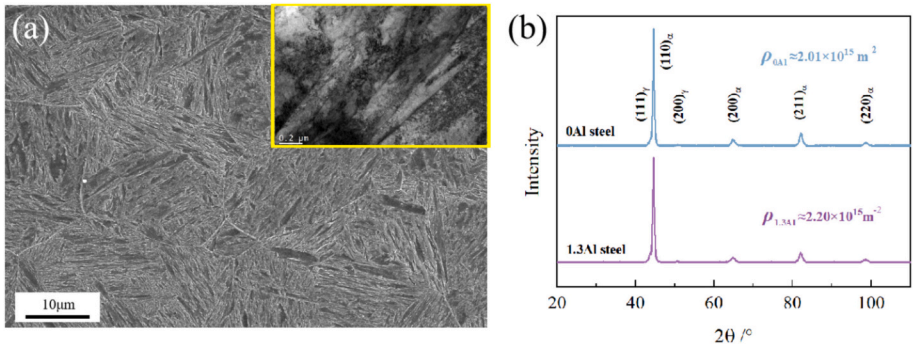


Fig. 2. (a) Microstructure of the 1.3Al steel; (b) X-ray diffraction (XRD) patterns of the 0Al and 1.3Al steels.

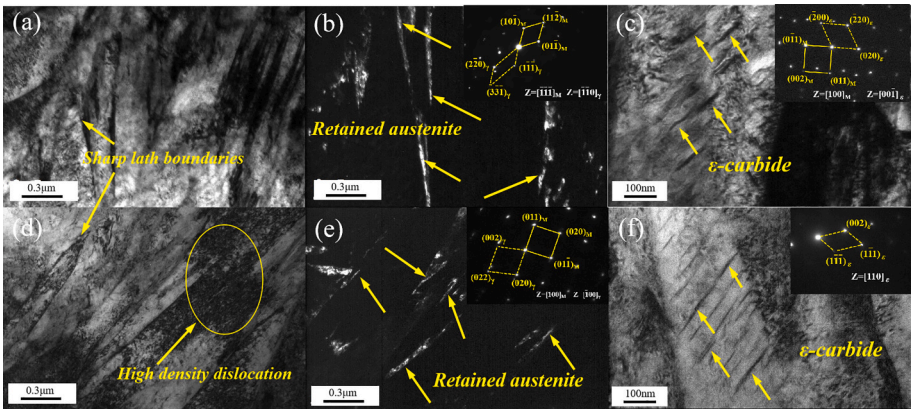


Fig. 3. Characterization of martensite, retained austenite, and ε-carbides in the two steels: (a-c) 0Al steel; (d-f) 1.3Al steel.

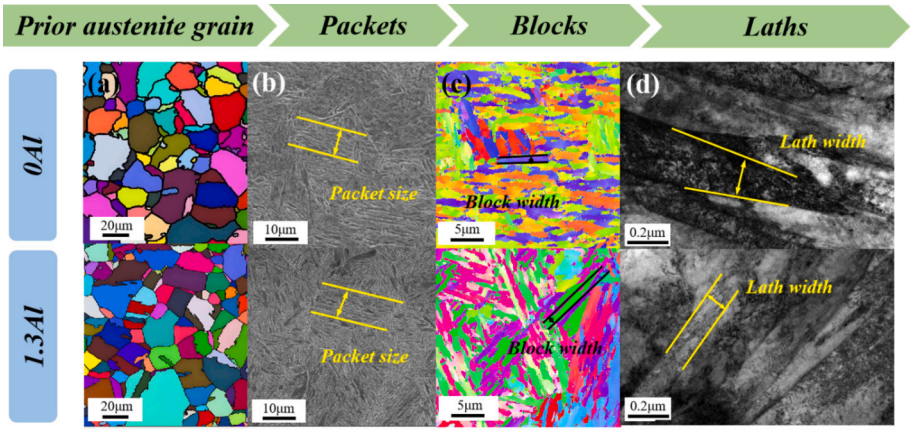


Fig. 4. Multilevel martensitic structures of the experimental steels: (a) prior austenite grains; (b) martensitic packets; (c) martensitic blocks; (d) martensitic laths.

Table 3
Multilevel martensitic structural characteristics of the two steels: average sizes of prior austenite grains (d_a), martensitic packets (d_p), martensitic blocks (d_b), and martensitic laths (d_l).

Steel	d_a/μm	d_p/μm	d_b/μm	d_l/μm
0Al	26.1 ± 9.4	3.4 ± 0.8	0.9 ± 0.2	0.24 ± 0.07
1.3Al	23.4 ± 5.9	3.5 ± 0.9	0.8 ± 0.2	0.23 ± 0.06

(Fig. 5(b)) and fast Fourier transform (FFT) analysis confirm these precipitates have a hexagonal close-packed (hcp) structure along the [110] zone axis. The proximity histogram (Fig. 5(c)) reveals that the precipitates mainly consist of segregated carbon, with no significant enrichment of other elements. Combined with TEM analysis, the precipitates are identified as ε-carbides [23–25]. Notably, slight Ni and Al enrichment is detected at their centers.

Near-spherical NiAl clusters were found to be remarkably uniformly distributed within the matrix, with a number density of 4.88×10^{24} m^{-3} and an equivalent radius of 0.57 nm. The FFT image in Fig. 5(e) reveals that the NiAl phase within the martensitic matrix exhibits a bcc structure along the [001] zone axis, with no discernible interface between the NiAl and the matrix, indicating a fully coherent relationship. The proximity histogram shown in Fig. 5(f) indicates that the atomic concentrations of Ni and Al at the cluster centers are approximately 13 at.%, significantly lower than the stoichiometric composition of a fully nucleated NiAl phase [17,26,27].This suggests that the NiAl clusters are still in the early stages of nucleation and exist as small clusters within the steel. Notably, slight segregation of Si, Mo, and Cr was also detected within the NiAl clusters.

Fig. 6 presents the distribution of ε -carbides and NiAl clusters within the matrix. Cross-sectional image (A region) shows that ε-carbides exhibit no enrichment of Ni, Al, or other elements. In the overall precipitate map, most NiAl clusters are uniformly dispersed throughout the matrix. Proximity analysis of the representative B region reveals that only Ni and Al concentrations differ between the clusters and the matrix, with other elements remaining constant, confirming a uniform distribution. In contrast, the C region exhibits a band-like distribution of NiAl clusters less than 5 nm in size., accompanied by co-segregation of Cr, Mo, and Si, as shown in Fig. 6C. Atomic concentration profiles indicate that Si, Cr, and Mo concentrations within these clusters increase significantly compared to the matrix, with Si rising from 4.21 at.% to 11.21 at.%, Cr from 1.65 at.% to 5.83 at.%, and Mo from 0.33 at.% to 1.35 at.%. This co-segregation suggests that the banded NiAl clusters may be associated with distinct interfaces within the martensitic matrix.

3.4. Fracture analysis

The load-deflection curves from the Charpy impact tests are shown in Fig. 7(a). Key points include the yield load (P_{GY}) and the maximum load (P_M). Crack initiation occurs at P_M, with the specimen remaining in the crack initiation stage prior to reaching P_M. The crack initiation energy (E_i) represents the material's resistance to crack formation, while the subsequent crack propagation energy (E_p) reflects the material's ability to resist crack growth, indicating the tendency toward ductile or brittle fracture behavior [28].

Fig. 7(a) shows that the 0Al and 1.3Al steels reached their peak loads (P_M) at deflections of 2.06 mm and 2.55 mm, respectively, followed by rapid load drops. Crack propagation accelerated at ~ 2.5 mm for the 0Al

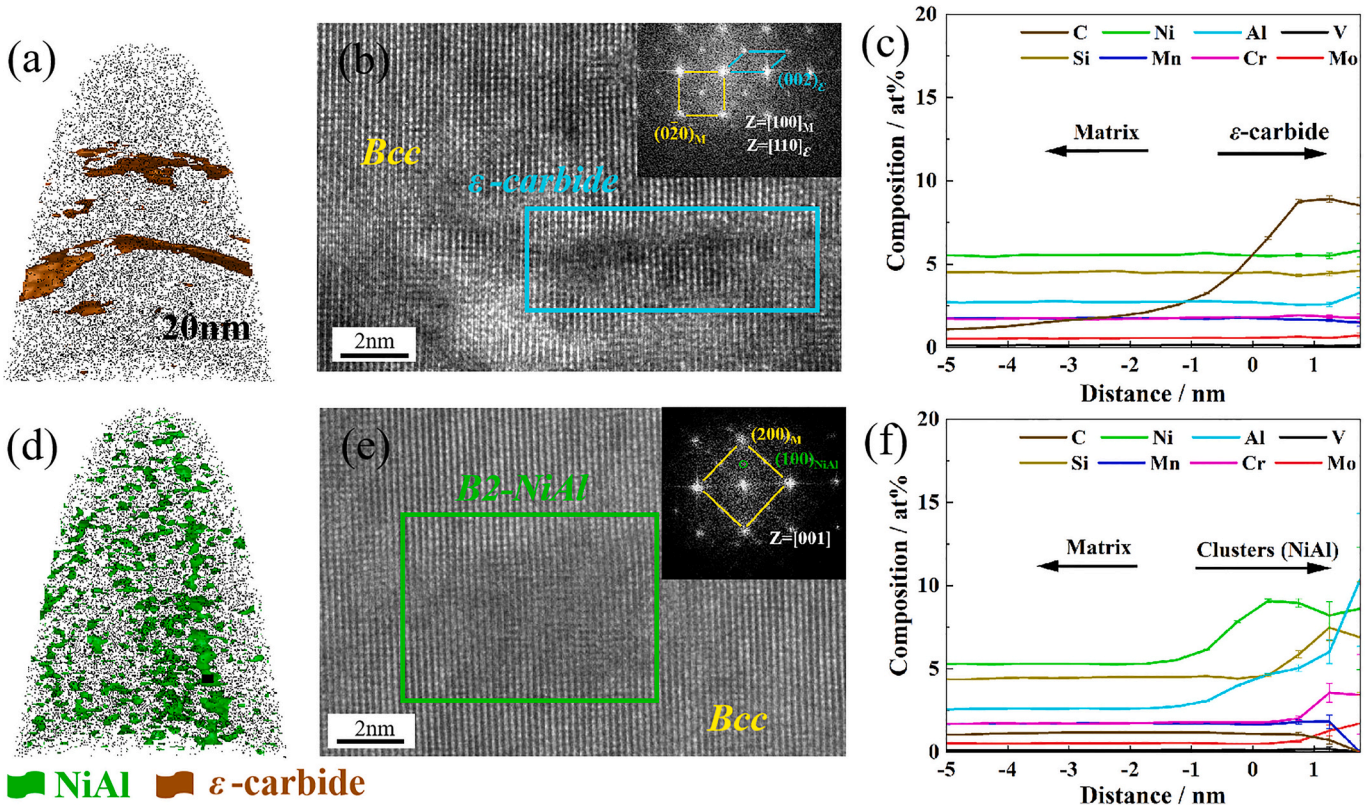


Fig. 5. Multiscale characterization of nanoparticles: all images were obtained from the 1.3Alsteel. (a,d) 5 at.% C isoconcentration surfaces showing ε-carbides and 10 at.% Ni + Al isoconcentration surfaces showing NiAl clusters; (b,e) TEM images and FFT patterns of ε-carbides and NiAl clusters; (c,f) Proximity histograms illustrating the compositional variations of ε-carbides and NiAl clusters.

steel and ~ 3 mm for the 1.3Al steel. Rapid fracture occurred near the specimen center without an obvious crack-arrest stage. Fig. 7(b) and Fig. 7(c) present the load-energy-displacement curves. Peak loads of 39.7 KN (0Al steel) and 38.6 KN (1.3Al steel) were achieved due to strain hardening, with substantial plastic deformation observed before P_M. The E_i and E_p were 45.0 J and 17.4 J for the 0Al steel, and 62.0 J and 21.0 J for the 1.3Al steel, respectively. Compared to the 0Al steel, the 1.3Al steel showed a 37.8 % higher E_i and a 20.7 % higher E_p.

Fig. 8 presents the impact fracture morphologies of the two steels, with observation areas marked in Fig. 8(a) and (d). White frames highlight the central regions and shear lips. Both the 0Al and 1.3Al steel exhibited pronounced shear lips, with calculated shear fracture areas of 36.8 % and 42.9 %. Notably, the notch region and initial fracture zone in the 1.3Al steel displayed significant plastic deformation. Numerous dimples near the crack initiation sites indicate ductile fracture via void growth and coalescence. In Fig. 8(b), primary dimples, fine secondary dimples, merged dimples (yellow regions), and elongated voids (green arrows) were observed. Fig. 8(c) shows finer secondary dimples and quasi-cleavage features (blue regions) along with extended dimple growth.

As shown in Fig. 8(e), the fracture surface of the 1.3Al steel also exhibits dominant ductile dimples and quasi-cleavage features. However, compared to Fig. 8(b), a notable increase in both the number and size of primary dimples is observed, accompanied by more prominent tearing ridges and greater overall surface roughness. The density of large and deep dimples is significantly higher in this region.

Fig. 9 shows the main crack paths and surrounding microstructures of the 0Al and 1.3Al steel. Fig. 9(a) and (c) display BC + IPF + GB maps, with prior austenite grain boundaries ($15° \leq \theta < 50°$) in yellow and packet or block boundaries ($\theta \geq 50°$) in blue [29,30]. In both steels, cracks propagated transgranularly, cutting across packets and deflecting at block boundaries, often causing bending or rupture of nearby blocks. In the 0Al steel, cracks similarly crossed packets and deviated along or through blocks.

The kernel average misorientation (KAM) calculated from EBSD data are shown in Fig. 9(b) and (d), with corresponding distributions on the right. The KAM maps display a θ range from 0° to 5°, reflecting the local strain distribution within the crystalline structure [31], where higher values correspond to larger strain. The 0Al steel exhibits an average KAM value of 1.16°, slightly lower than the 1.31° measured for the 1.3Al steel. However, as observed in Fig. 9(b), the region surrounding the main crack in the 0Al steel is dominated by red and yellow regions, while significant blue undeformed areas are observed in the adjacent matrix, suggesting an inhomogeneous strain distribution. In contrast, Fig. 9(d) reveals a relatively uniform distribution of KAM values uniformly across both the impact fracture surface and the surrounding matrix in the 1.3Al steel, indicating a more homogeneous strain distribution.

Cross-sectional microstructures of fracture toughness specimens were analyzed (Fig. 10). In Fig. 10(a) and (b), cyan arrows trace the main crack paths. The 1.3Al steel showed frequent crack deflections, while the 0Al steel exhibited straighter propagation with minimal deviations, suggesting faster crack growth.

In the 1.3Al steel, secondary cracks and voids measuring 5–10 μm were observed 8–18 μm below the main crack as shown in Fig. 10(d), with smaller voids closer to the matrix. Fine cracks interconnected these voids, indicating coalescence into larger secondary cracks. However, the 0Al steel displayed few voids and negligible secondary cracking. Main cracks occasionally deflected but rapidly blunted upon penetrating the matrix [32].

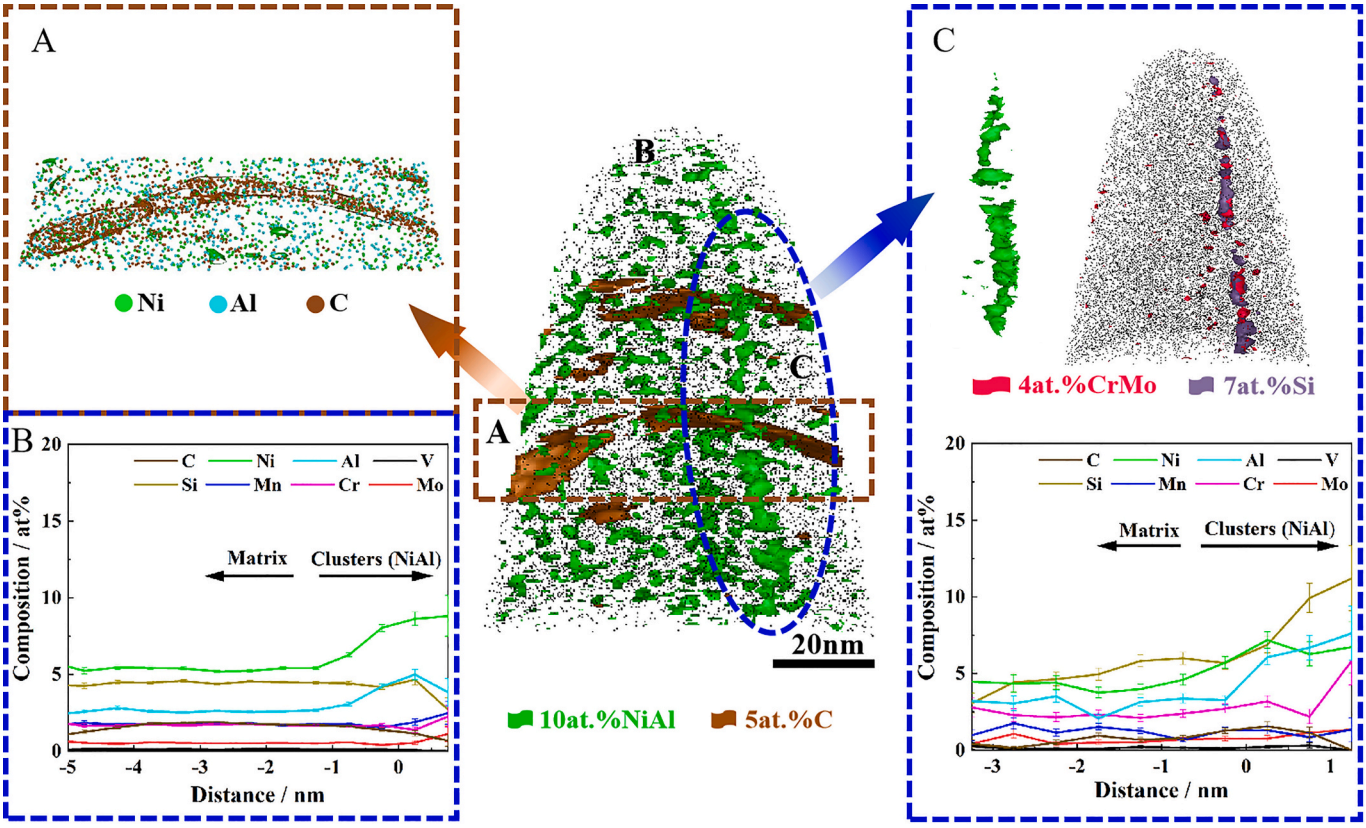


Fig. 6. APT characterization of the 1.3Al steel: (A) atomic distribution of Ni, Al, and C across an ε-carbide cross-section; (B) proximity histogram showing compositional variations of dispersed NiAl clusters; (C) 4 at.% Cr + Mo and 7 at.% Si isoconcentration surfaces and proximity histogram of segregated NiAl clusters.

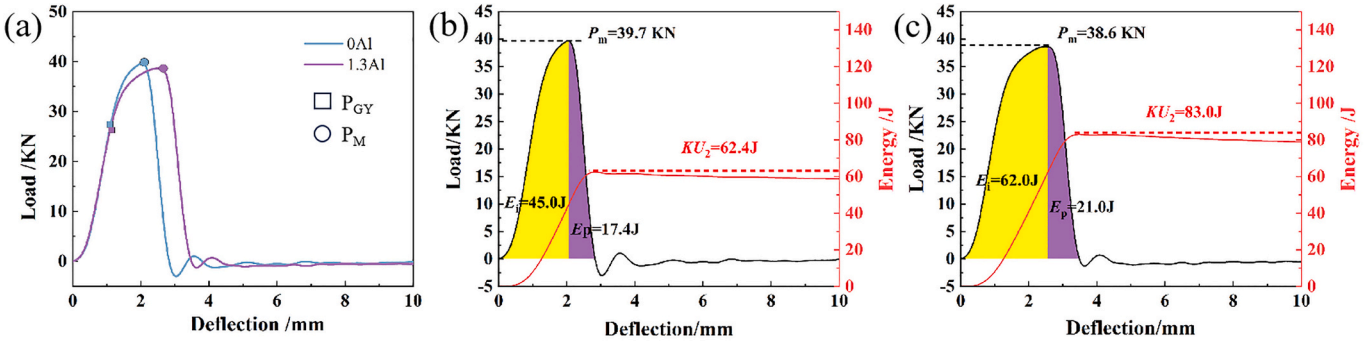


Fig. 7. Load-deflection curves of the experimental steels: (a) comparison between the two steels; (b) 0Al steel; (c) 1.3Al steel.

4. Discussion

4.1. Precipitation behavior

APT and TEM analyses (Figs 3, 5 and 6) reveal that, in contrast to the 0Al steel, the 1.3Al steel contains not only ε-carbides formed during low-temperature tempering at 180 °C, but also uniformly dispersed NiAl clusters with an extremely high number density of 4.88×10^{24} m^{-3}. The hcp-structured ε-carbides measure approximately 100–500 nm in length and 2–4 nm in thickness, and are regularly aligned along the martensitic laths. In contrast, B2-ordered NiAl clusters with an average radius of approximately 0.57 nm are uniformly dispersed throughout the matrix.

APT isoconcentration maps (Fig. 5(c) and Fig. 6A) reveal that the central region of the ε-carbides contains a carbon concentration as high as 8.91 at.%, while the concentrations of Ni and Al are ~ 5.50 at.% and ~ 2.60 at.%, respectively. As shown in Fig. 5(c), the Ni and Al contents remain essentially constant in this region. In contrast, the region surrounding NiAl clusters consistently maintains a carbon concentration of ~ 1.0 at.% (Fig. 6B and C). These findings indicate that ε-carbides and NiAl clusters nucleate and grow independently during tempering, with no observable elemental overlap or competitive partitioning between C and Ni/Al.

The lattice parameter of NiAl (~2.882 nm) closely matches that of the martensitic matrix (~ 2.866 nm), resulting in a lattice mismatch of only ~ 0.5 % [18,33]. The presence of Mo and Cr atoms expands the matrix lattice, thereby further reducing the lattice mismatch and lowering the nucleation barrier [34]. which promotes the extremely uniform precipitation of NiAl clusters.

Fig. 6C reveals that, while most NiAl clusters are uniformly distributed within the martensitic matrix, a fraction preferentially precipitates along Si- and Cr-enriched banded zones. In these localized regions, the Si content rises significantly from 4.21 at.% in the matrix to 11.21 at.%

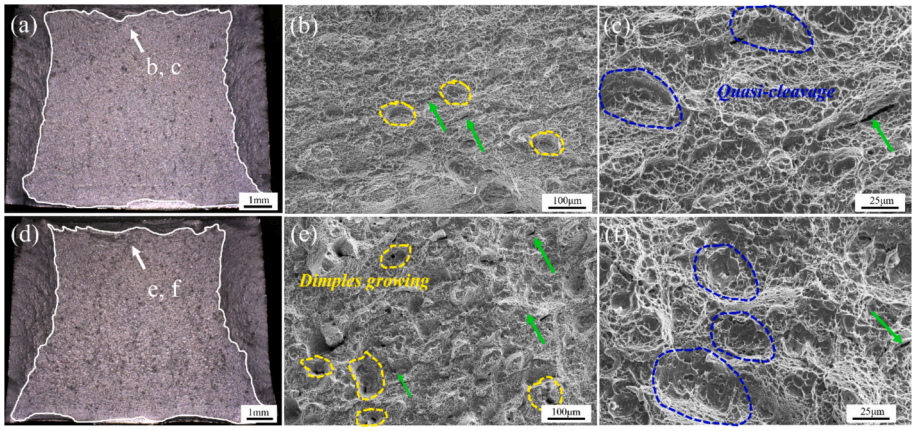


Fig. 8. Impact fracture morphologies: (a) macroscopic fracture surface of the 0Al steel; (b, c) microscopic features from the regions indicated by arrows in (a); (d) macroscopic fracture surface of the 1.3Al steel; (e, f) microscopic features from the regions indicated by arrows in (d).

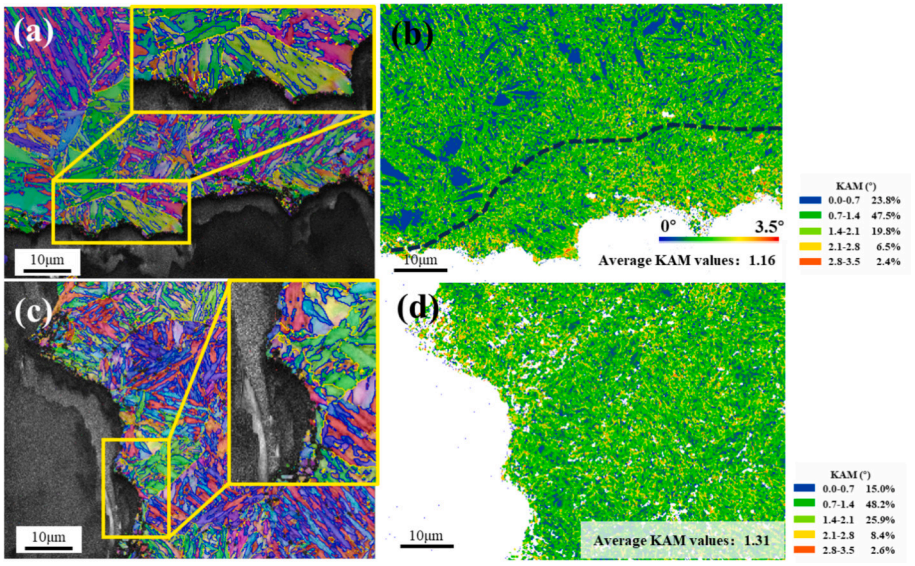


Fig. 9. Cross-sectional features of the impact fractures in the 0Al and 1.3Al steels: (a, b)0Al steel; (c, d) 1.3Al steel. (b, d) are magnified views of the regions marked in (a, c), respectively. (a, c) show BC + IPF + GB maps near the main cracks; (b, d) show corresponding KAM maps near the main cracks.

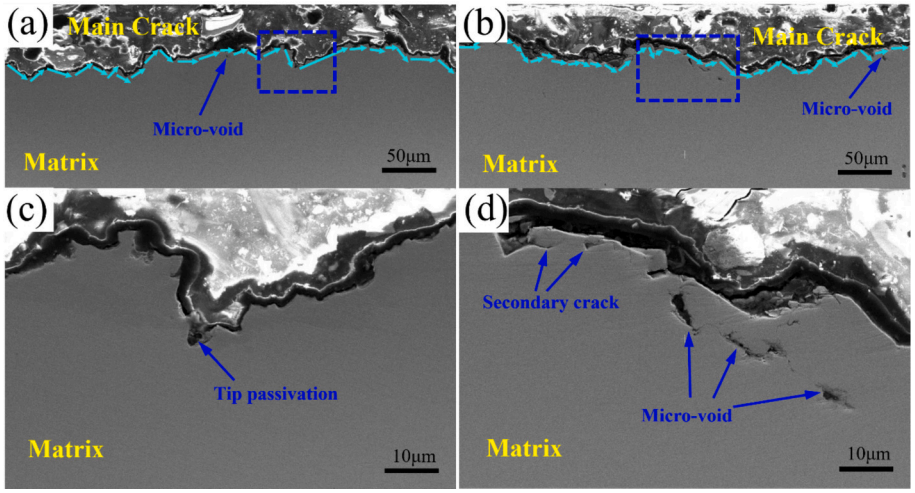


Fig. 10. The fracture vertical-section microstructures of fracture toughness specimens: (a, c) for 0Al steel specimen; (b, d) for 1.3Al steel specimen; (c), (d) are the magnified images marked by the blue box in (a), (b). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

indicating solute segregation. The NiAl clusters in this region exhibit slightly larger sizes than the matrix-dispersed counterparts, while still retaining nanoscale dimensions. Previous studies have reported that NiAl tends to nucleate at defects such as dislocations [29] and boundaries [19,35]. However, the absence of carbon segregation in the banded region observed in Fig. 6C suggests that it does not correspond to a typical phase boundary, but is more likely associated with non-dislocation intragranular defects [29,36]. In conventional low alloy steels, Si has been reported to stabilize the ε phase and delay its transformation to cementite during tempering at elevated temperatures (250–350 °C), thereby mitigating temper embrittlement [37]. However, under lower tempering conditions (<250 °C), no significant variation in Si content is typically observed, nor is it known to induce other reactions [38]. However, in the present study, Si enrichment was observed around NiAl clusters after tempering at 180 °C (Fig. 6C), indicating an interaction between Si and NiAl. These Si-enriched regions are considered to provide favorable nucleation sites for NiAl clusters, thereby facilitating their precipitation.

4.2. Dislocation regulation by NiAl cluster

To investigate the interaction between NiAl clusters and dislocations, statistical analysis was conducted on multiple fields near the tensile fracture surface (Supplementary Fig. S2 and Fig. S3). Under sustained stress, a large number of dislocation cells are formed within the martensitic laths. Compared with the reference steel, the dislocation distribution in the 1.3Al steel appears more homogeneous.

Despite being sheared by mobile dislocations during deformation [39], the ultrafine and densely distributed NiAl clusters significantly hinder dislocation motion by increasing cluster density and reducing the dislocations mean free path, thereby enhancing lattice friction and promoting a more uniform dislocation structure [40]. The distribution of KAM values reflected the dislocation density of the specimen surface [41]. Fig. 9 demonstrates that the uniform KAM distribution in the 1.3Al steel facilitates effective stress transfer to the interior of the matrix during fracture, enabling coordinated deformation and mitigating stress concentration near the main crack. The KAM value in the matrix of 1.3Al and 2 h steels is low (0.79, as shown in Fig. S5), but near the fracture surface, it increases to 1.31, indicating that the elevated KAM values are not intrinsic to the alloy, but rather a result of stress dispersion near the crack. This confirms that NiAl clusters facilitate the formation of spatially uniform dislocation structures through cooperative interactions with dislocations. This suppresses entanglement and precipitate-induced pile-ups, mitigating local stress concentrations [42]. While the 1.3Al steel exhibits a slightly higher dislocation density of $\sim 2.20 \times 10^{15}$ m^{-2} compared to $\sim 2.01 \times 10^{15}$ m^{-2} in the 0Al steel, its improved dislocation homogeneity contributes to enhanced fracture toughness.

4.3. Toughening mechanisms

Toughening mechanisms are typically governed by the synergistic effect of microstructural features in regulating stress distribution and accommodating plastic deformation during crack initiation and propagation [43]. Based on the result of Fig. 7-Fig. 10, the toughening mechanisms in 1.3Al steel are analyzed in three aspects:

Prior to P_M, both the 0Al and 1.3Al steels exhibited significant plastic deformation, with strain hardening raising P_M to ~ 40 KN (Fig. 7). Stable crack propagation followed, with the 1.3Al steel absorbing more impact energy, consistent with fractographic evidence (Fig. 8). Fracture in the 1.3Al steel was dominated by ductile mechanisms involving the growth and coalescence of large and small dimples. Under applied stress, numerous voids are generated within the matrix, which expand and coalesce into larger dimples. This void formation and growth dissipate a substantial portion of the impact energy during fracture [44].

Secondly, the higher E_i observed in the 1.3Al steel (Fig. 7) reflects enhanced plastic deformability and crack resistance. The uniformly distributed dislocation network effectively constrains the formation of pile-ups during the interaction of dislocations, thereby suppressing localized stress intensification [35]. During fracture, the shearable and coherent nature of NiAl clusters facilitates progressive strain delocalization into the surrounding matrix, enabling extensive and coordinated plastic deformation. This redistribution of strain not only elevates the energy threshold required for crack initiation, but also delays the onset of critical stress concentrations, thereby enhancing the material's ability to sustain higher applied loads without premature failure [35].

Additionally, Dense packet boundaries and dislocations (low-angle boundaries) act as barriers to dislocation motion [27].As shown in Fig. 10, the 1.3Al steel exhibits frequent crack deflections and voids with a length of ~10 μm near the fracture surface due to localized strain accumulation [45]. The coalescence of voids can initiate secondary cracks within the matrix, leading to deflection of the main crack and enhanced crack tip blunting [46,47]. This crack path deviation increases the propagation distance and promotes greater energy dissipation [48]. Such a toughening effect is particularly evident in the 1.3Al steel, where E_p exceeds that of the 0Al counterpart by 4.6 J (Fig. 7), confirming the effectiveness of this mechanism. The less pronounced increase in E_p relative to E_i provides additional evidence that the improvement in toughness is predominantly governed by the structural uniformity imparted by the dispersion of NiAl clusters, rather than by the 0.8 % increase in retained austenite content [5,49,50].

Fig. 11 schematically depicts the crack propagation mechanisms operative during impact fracture in both low alloy ultrahigh strength steels. In the 1.3Al steel, uniformly distributed NiAl clusters and ε-carbides interact with dislocations in distinct ways: ε-carbides promote dislocation cell formation through bypassing [51], while closely spaced NiAl clusters are directly sheared, promoting a more uniform spatial distribution. This refined dislocation network suppresses localized strain accumulation and raises the energy barrier for crack initiation. At high stress levels, the crack path is further deflected by NiAl-induced perturbations and disrupted by secondary cracks initiated via voids coalescence. These combined effects increase crack tortuosity and retard propagation, resulting in greater energy absorption. Collectively, the hierarchical dislocation-obstacle architecture enables a favorable strength-toughness synergy by dispersing strain, delaying failure, and impeding crack advancement.

5. Conclusions

A novel low alloy ultrahigh strength steel was developed by introducing coherent NiAl clusters, achieving superior strength-toughness synergy. The main findings are:

1. High-density coherent NiAl clusters ($\sim 4.88 \times 10^{24}$ m^{-3}) with high nucleation rates and uniform dispersion were successfully introduced. A small fraction of NiAl clusters exhibits a specific interaction with Si, which increases the availability of heterogeneous nucleation sites.
2. NiAl clusters optimized dislocation distributions, thereby enhancing strain hardening, plastic deformation capacity, and delaying crack initiation.
3. Toughening mechanisms including large dimple growth, extensive plastic deformation, secondary crack formation via voids coalescence, and frequent crack deflection significantly enhanced impact energy absorption.
4. The NiAl-cluster-strengthened steel exhibited a *YS* of 1272 ± 52 MPa, a *UTS* of 1876 ± 25 MPa, a *TE* of 13.7 ± 0.4 %, a KU_2 of 80 ± 3 J, and a K_{IC} of 108 MPa·m$^{1/2}$, demonstrating excellent strength-toughness balance.

Overall, the uniform precipitation of NiAl clusters offers an effective pathway for enhancing the strength-toughness synergy of low alloy

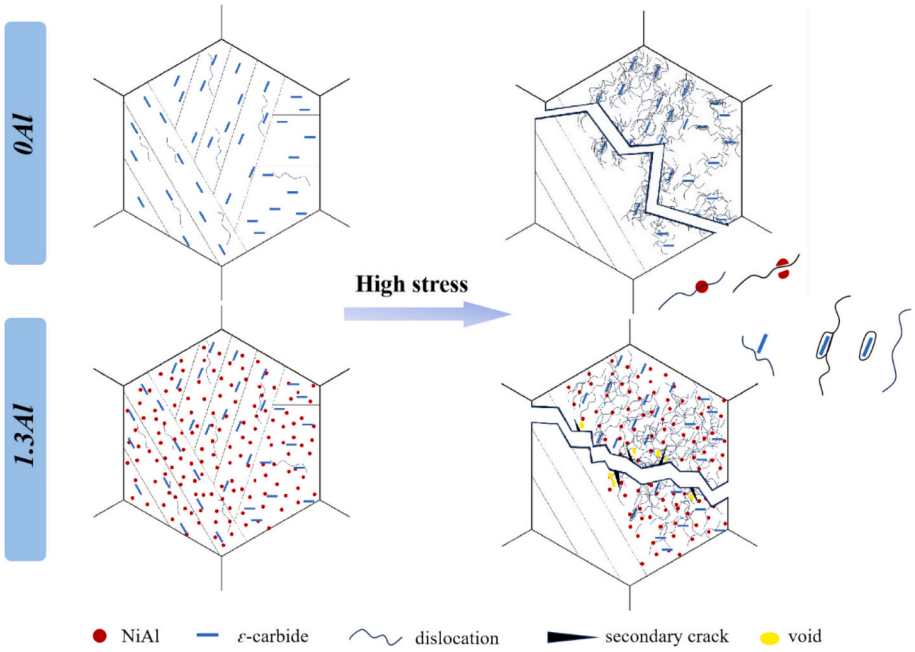


Fig. 11. Schematic illustration of the crack propagation behavior during impact fracture.

martensitic steels, providing insights for the design of next-generation ultrahigh strength materials.

CRediT authorship contribution statement

Xiaohan Duan: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Xin Cao:** Formal analysis. **Yangxin Wang:** Writing – original draft, Validation, Methodology. **Hongshan Zhao:** Formal analysis. **Chundong Hu:** Writing – original draft, Validation, Methodology. **Han Dong:** Writing – original draft, Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The financial support from the National Natural Science Foundation of China (Key Program): 52031004.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matchar.2025.115480.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

[1] D.B. Garcia, R. Forman, D. Shindo, Experimental evaluation of fatigue crack initiation from corroded hemispherical notches in aerospace structural materials, Fatigue Fract. Eng. Mater. Struct. 35 (2) (2012) 122–140.
[2] P. Skubisz, J. Sinczak, Properties of direct-quenched aircraft forged component made of ultrahigh-strength steel 300M, Aircr. Eng. Aerosp. Technol. 90 (5) (2018) 713–719.
[3] B.M. Leister, J.N. DuPont, Development of a continuous cooling transformation diagram for Eglin steel, Mater. Sci. Technol. 31 (12) (2015) 1425–1432.
[4] W.T. Xu, L. Xie, X.Y. Liu, J.N. Wang, Y.X. Xu, M.T. He, K.J. Hu, C. Liu, W. Yu, The fabrication of ultrahigh-strength steel with a nanolath structure via quenching–partitioning–tempering, Materials 17 (5) (2024) 1161.
[5] T. Wang, J.J. Sheng, Y.H. Deng, Y.X. Wang, C.D. Hu, H. Dong, Regulation of nanosized retained-austenite morphology making low alloy ultrahigh strength steel tough, J. Mater. Res. Technol. 35 (2025) 3563–3572.
[6] G. Mandal, C. Roy, S.K. Ghosh, S. Chatterjee, Structure-property relationship in a 2 GPa grade micro-alloyed ultrahigh strength steel, J. Alloys Compd. 705 (2017) 817–827.
[7] M. Abdelwahed, S. Bengtsson, M. Boniardi, A. Casaroli, R. Casati, M. Vedani, An investigation on the plane-strain fracture toughness of a water atomized 4130 low-alloy steel processed by laser powder bed fusion, Mater. Sci. Eng. A 855 (2022) 143941.
[8] F.G. Liu, W.J. Zhang, X. Lin, C.P. Huang, F.C. Liu, W.D. Huang, P.F. Wang, X. Li, Effect of isothermal temperature on bainite transformation, microstructure and mechanical properties of LSFed 300M steel, Mater. Today Commun. 25 (2020) 101452.
[9] D. Satyanarayana, G. Malakondaiah, D. Sarma, Characterization of the age-hardening behavior of a precipitation-hardenable austenitic steel, Mater. Charact. 47 (1) (2001) 61–65.
[10] H.W. Luo, X.H. Wang, Z.B. Liu, Z.Y. Yang, Influence of refined hierarchical martensitic microstructures on yield strength and impact toughness of ultra-high strength stainless steel, J. Mater. Sci. Technol. 51 (2020) 130–136.
[11] C. Zhao, C. Zhang, W.Q. Cao, Z.G. Yang, Variation in retained austenite content and mechanical properties of 0.2 C–7Mn steel after intercritical annealing, Int. J. Miner. Metall. Mater. 23 (2016) 161–167.
[12] D. Jain, D. Isheim, A.H. Hunter, D.N. Seidman, Multicomponent high-strength low-alloy steel precipitation-strengthened by sub-nanometric Cu precipitates and M_2C carbides, Metall. Mater. Trans. A 47 (2016) 3860–3872.
[13] Z.P. Xiong, I. Timokhina, E. Pereloma, Clustering, nano-scale precipitation and strengthening of steels, Prog. Mater. Sci. 118 (2021) 100764.
[14] C.A. Stewart, R.W. Fonda, K.E. Knipling, Mn-stabilized austenitic steel strengthened by nano-scale β-NiAl (B2), FCC-cu, and carbides via ICME design, Scr. Mater. 200 (2021) 113903.
[15] C.A. Stewart, E.A. Antillon, M. Sudmanns, J.A. El-Awady, K.E. Knipling, P. G. Callahan, D.J. Rowenhorst, R.W. Fonda, Predicting yield stress in a nano-precipitate strengthened austenitic steel by integrating multi length-scale simulations and experiments, Acta Mater. 272 (2024) 119918.
[16] H.F. Zhu, Z.P. Xiong, J.W. Mao, X.W. Cheng, Achieving ultra-high strength, good toughness and cost reduction in secondary hardening steel via dual precipitation, Mater. Charact. 223 (2025) 114869.
[17] Y.H. Gao, S.Z. Liu, X.B. Hu, Q.Q. Ren, Y. Li, V.P. Dravid, C.X. Wang, A novel low cost 2000 MPa grade ultra-high strength steel with balanced strength and toughness, Mater. Sci. Eng. A 759 (2019) 298–302.
[18] Z.K. Teng, G. Ghosh, M.K. Miller, S. Huang, B. Clausen, D.W. Brown, P.K. Liaw, Neutron-diffraction study and modeling of the lattice parameters of a NiAl-precipitate-strengthened Fe-based alloy, Acta Mater. 60 (13–14) (2012) 5362–5369.

[19] B. Peng, H. Wang, M. Zhou, R. Zhang, X.T. Li, J. Wu, G.P. Xu, L.X. Zhou, H.T. Dong, S.H. Jiang, Ultrastrong metallic materials via minimal lattice misfit and strong ordering effect, Sci. China Mater. 66 (11) (2023) 4182–4188.
[20] G.K. Williamson, W.H. Hall, X-ray line broadening from filed aluminium and wolfram, Acta Metall. 1 (1) (1953) 22–31.
[21] G.K. Williamson, R.E. Smallman, The use of Fourier analysis in the interpretation of X-ray line broadening from cold-worked iron and molybdenum, Acta Crystallogr. 7 (8–9) (1954) 574–581.
[22] Z.B. Jiao, J.H. Luan, M.K. Miller, C.Y. Yu, Y. Liu, C.T. Liu, Precipitate transformation from NiAl-type to Ni2AlMn-type and its influence on the mechanical properties of high-strength steels, Acta Mater. 110 (2016) 31–43.
[23] C.B. Yan, W.H. Ding, L. Liang, X.H. Liu, Y. Yang, H.B. Wang, S.L. Xiao, W.Z. Yang, The behavior of ε-carbides and their influence on residual stress, Steel Res. Int. 95 (7) (2024) 2400014.
[24] W. Song, J. Von Appen, P. Choi, R. Dronskowski, D. Raabe, W. Bleck, Atomic-scale investigation of ε and θ precipitates in bainite in 100Cr6 bearing steel by atom probe tomography and ab initio calculations, Acta Mater. 61 (20) (2013) 7582–7590.
[25] J. Takahashi, K. Kawakami, Y. Sakiyama, T. Ohmura, Atomic-scale observation of hydrogen trap sites in bainite–austenite dual-phase steel by APT, Mater Charact 178 (2021) 111282.
[26] S.H. Jiang, H. Wang, Y. Wu, X.J. Liu, H.H. Chen, M.J. Yao, B. Gault, D. Ponge, D. Raabe, A. Hirata, Ultrastrong steel via minimal lattice misfit and high-density nanoprecipitation, Nature 544 (7651) (2017) 460–464.
[27] T.Q. Liu, Z.X. Cao, H. Wang, G.L. Wu, J.J. Jin, W.Q. Cao, A new 2.4 GPa extra-high strength steel with good ductility and high toughness designed by synergistic strengthening of nano-particles and high-density dislocations, Scr. Mater. 178 (2020) 285–289.
[28] E. Tkachev, S. Borisov, A. Belyakov, T. Kniaziuk, O. Vagina, S. Gaidar, R. Kaibyshev, Effect of quenching and tempering on structure and mechanical properties of a low-alloy 0.25 C steel, Materials Science Engineering A 868 (2023) 144757.
[29] X.Y. Zhang, J.L. Wang, S.Y. Liu, L. Yan, C.H. Song, H. Yu, Developing NiAl-strengthened HSLA steels by controlling nanoscale precipitation and high-angle boundaries, Mater. Sci. Eng. A 861 (2022) 144355.
[30] Z.K. Ai, H. Teng, C. Ge, H.S. Zhao, B. Wang, H. Dong, Influence of multilevel martensitic microstructures and carbide on the low-temperature toughness and fracture behavior of MnB steels, Mater. Sci. Eng. A 925 (2025) 147909.
[31] T. Zhou, H. Yu, S.Y. Wang, Effect of microstructural types on toughness and microstructural optimization of ultra-heavy steel plate: EBSD analysis and microscopic fracture mechanism, Mater. Sci. Eng. A. 658 (2016) 150–158.
[32] F. Wang, D.S. Qian, L. Hua, L.C. Xie, Tempering response and improved mechanical properties in secondary hardened steel by introducing an optimized austempering process, Mater. Sci. Eng. A 807 (2021) 140895.
[33] Z.K. Teng, M.K. Miller, G. Ghosh, C.T. Liu, S. Huang, K.F. Russell, M.E. Fine, P. K. Liaw, Characterization of nanoscale NiAl-type precipitates in a ferritic steel by electron microscopy and atom probe tomography, Scr. Mater. 63 (1) (2010) 61–64.
[34] Z.B. Jiao, J.H. Luan, Z.W. Zhang, M.K. Miller, C.T. Liu, High-strength steels hardened mainly by nanoscale NiAl precipitates, Scr. Mater. 87 (2014) 45–48.
[35] X.Y. Zhang, J.L. Wang, T. Zhou, L. Yan, H. Yu, Impact toughness and fracture propagation mechanism of NiAl precipitation-strengthened HSLA steels, Mater. Des. 241 (2024) 112927.
[36] D.Y. Huang, J.C. Yan, X.W. Zuo, Co-precipitation kinetics, microstructural evolution and interfacial segregation in multicomponent nano-precipitated steels, Mater. Charact. 155 (2019) 109786.
[37] B. Kim, C.d. Celada, D. San Martín, T. Sourmail, D.d.C. Rivera, The effect of silicon on the nanoprecipitation of cementite, Acta Mater. 61 (18) (2013) 6983–6992.
[38] A.J. Clarke, M.K. Miller, R.D. Field, D.R. Coughlin, P.J. Gibbs, K.D. Clarke, D. J. Alexander, K.A. Powers, P.A. Papin, Atomic and nanoscale chemical and structural changes in quenched and tempered 4340 steel, Acta Mater. 77 (2014) 17–27.
[39] R. Schroll, V. Vitek, P. Gumbsch, Core properties and motion of dislocations in NiAl, Acta Mater. 46 (3) (1998) 903–918.
[40] T. Edahiro, K. Kouzai, H.Y. Yasuda, Mechanical properties and hardening mechanism of Fe–Al–Ni single crystals containing NiAl precipitates, Acta Mater. 61 (5) (2013) 1716–1725.
[41] L. Liu, W. Huang, M. Ruan, Z. Chen, Effects of temperatures on microstructure evolution and deformation behavior of Fe–32Ni by in-situ EBSD, Mater. Sci. Eng. A 875 (2023) 145097.
[42] R. Kositski, D. Mordehai, Role of dislocation pile-ups in nucleation-controlled size-dependent strength of Fe nanowires, Acta Mater. 136 (2017) 190–201.
[43] R.O. Ritchie, The conflicts between strength and toughness, Nat. Mater. 10 (11) (2011) 817–822.
[44] K. Srinivasan, Y. Huang, O. Kolednik, T. Siegmund, The size dependence of micro-toughness in ductile fracture, J. Mech. Phys. Solids 56 (8) (2008) 2707–2726.
[45] A. Das, Stress/strain induced void? Arch. Comput. Methods Eng. 28 (3) (2021) 1795–1852.
[46] A.R. Chintha, K. Valtonen, V.T. Kuokkala, S. Kundu, M.J. Peet, H.K.D. H. Bhadeshia, Role of fracture toughness in impact-abrasion wear, Wear 428 (2019) 430–437.
[47] A. Das, H.W. Viehrig, F. Bergner, C. Heintze, E. Altstadt, J. Hoffmann, Effect of microstructural anisotropy on fracture toughness of hot rolled 13Cr ODS steel–the role of primary and secondary cracking, J. Nucl. Mater. 491 (2017) 83–93.
[48] R.O. Ritchie, J.F. Knott, J. Rice, On the relationship between critical tensile stress and fracture toughness in mild steel, J. Mech. Phys. Solids 21 (6) (1973) 395–410.
[49] W.H. Peng, Y.C. Wu, R.M. Geng, Y.X. Cao, S. Han, Y. Li, C.X. Wang, L.J. Zheng, Microstructures and mechanical properties of novel 2.3 GPa secondary hardening steels with different Ni contents, J. Mater. Res. Technol. 35 (2025) 675–684.
[50] G. Mishra, M.K. Bhatt, K.A. Anand, S. Biswal, S.M. Hasan, S. Bagui, A. Ayyandurai, S. Samanta, A. Ghosh, A. Karmakar, Strength and toughness balance in 7% Ni steel by formation epsilon martensite, retained austenite and low matrix strain, Acta Mater. 36 (2024) 102183.
[51] Y. Kawahara, K. Kaneko, H. Sawada, J. Takahashi, Transition from carbon clusters to ε, θ-carbides in a quenched and aged low-carbon ferritic steel, Acta Mater. 252 (2023) 118919.